

Age Matters: The Effects of Trade Liberalization on Firms over the Life Cycle^{*}

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Abstract

This paper shows that firms' responses to trade shocks depend on their age. Focusing on India's trade liberalization during the 1990s, we find that reductions in input tariffs raise quantities and sales more for younger firms than for older firms. These age differences are not driven by selection effects and are not explained by age-varying observables such as firm size or productivity. To interpret the evidence, we develop a firm life-cycle model in which adjustment costs lead young firms to rely more heavily on flexible intermediate inputs as they grow toward mature scale. The model shows that this dynamic input mix makes input-cost reductions especially valuable early in the firm life cycle. The results highlight firm age as a distinct dimension of heterogeneity in the transmission of trade liberalization.

Keywords: Firm life cycle; Trade liberalization; Firm dynamics.

JEL Codes: F14, F61, O24.

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1 Introduction

Trade liberalization changes the economic environment in which firms operate. It exposes domestic producers to foreign competition, lowers the cost of imported inputs, and can alter firms' incentives to invest, expand, upgrade, or exit. A large literature shows that these effects are heterogeneous across firms, with productivity, exporting status, input sourcing, innovation, and financial constraints shaping how firms adjust to trade shocks.¹ This paper studies a dimension of heterogeneity that has received less attention in this context: firm age. Firms are not static production units and, over time, they accumulate experience, organizational routines, managerial capacity, customer and supplier relationships, and other intangible assets that are difficult to observe directly but central to firm performance. Firm age is therefore a valuable state variable that is both easy to measure and economically informative: it can summarize life-cycle processes that may otherwise be missed when heterogeneity is measured only through contemporaneous size or productivity.²

In this paper, we ask whether trade liberalization affects young and old firms differently. The question is important for both positive and normative reasons. From a positive perspective, age may shape how firms absorb trade shocks because young firms differ from mature firms not only in scale and productivity, but also in their stage of organizational development. Younger firms may be more flexible, more volatile, and further from their desired scale; older firms may have more established production processes, thicker market relationships, and more accumulated know-how. These differences are key in models and empirical studies of firm dynamics, where age is closely related to growth, exit, volatility, and adjustment behavior.³ From a policy perspective, the answer matters because young firms are often viewed as important contributors to future growth, job creation, and reallocation ([Haltiwanger et al., 2013](#)). If trade liberalization disproportionately harms

¹See [Clerides et al. \(1998\)](#), [Bernard and Jensen \(1999\)](#), [Eaton and Kortum \(2002\)](#), [Melitz \(2003\)](#), [Manova \(2013\)](#), [Halpern et al. \(2015\)](#) and [Bloom et al. \(2016\)](#).

²This interpretation is consistent with evidence that firms' life-cycle growth reflects not only contemporaneous productivity, but also accumulated demand, selection, organizational capabilities, and barriers to scaling ([Foster et al., 2008](#); [Hsieh and Klenow, 2014](#); [Akcigit et al., 2021](#)).

³A non-comprehensive list of papers showing this includes [Jovanovic \(1982\)](#), [Evans \(1987\)](#), [Hopenhayn \(1992\)](#), [Caves \(1998\)](#).

young firms, it may weaken the pool of future high-growth producers. If instead young firms benefit especially from cheaper inputs, trade reform may increase the value of entry and accelerate firm growth early in the life cycle.

We study this question in the context of India’s trade liberalization during the 1990s. The Indian reform provides a well-known setting in which tariff reductions were large, broad-based, and plausibly exogenous to firm-level outcomes ([Topalova, 2010](#); [Topalova and Khandelwal, 2011](#)). Following the identification strategy used in the literature on India’s trade reform, we distinguish between output tariffs, which affect the degree of import competition faced by firms in their product markets, and input tariffs, which affect the cost of intermediate goods used in production ([De Loecker et al., 2016](#)). This distinction is important because the two channels may have different implications over the firm life cycle. Output tariff reductions increase competitive pressure, while input tariff reductions lower production costs and may be especially valuable for firms still expanding and adjusting their input mix.

We exploit both firm and product-level data from PROWESS and plant-level data from the Annual Survey of Industries, to estimate whether tariff changes affect firms differently by age. Our first result is that reductions in output tariffs reduce firm sales and prices, consistent with stronger import competition, but these effects do not vary systematically with firm age. Young firms do not appear to be disproportionately harmed by increased foreign competition. Our second result is that reductions in input tariffs raise quantities and sales for young firms, but not for old firms. Firms early in their life cycle respond more strongly to cheaper inputs than mature firms do. We find no comparable effect on prices, implying that the differential effect on sales operates through quantities. To illustrate the differential age effect, our results imply that even though firms younger than 20 years account for 45% of sales in our data, they account for the entire sales response to input liberalization.

We then examine whether this age gradient reflects other observable characteristics that vary over the firm life cycle. Young firms are typically smaller, less productive, less diversified, and may differ in their export behavior, imported input use, markups,

capital intensity, financial constraints, and output quality. We therefore test whether the differential response to input tariff reductions is explained by these characteristics. We find that it is not, as controlling flexibly for these dimensions leaves the age effect largely intact. We also examine whether selection drives the results: older firms may appear less responsive simply because more responsive firms exit before reaching old age. A balanced-panel exercise shows similar results compared to the baseline, suggesting that the age effect is not a selection artifact.

To interpret the evidence, we develop a firm life-cycle model with capital adjustment costs and flexible intermediate input use. The model focuses on physical capital as a measurable state variable that proxies for the broader process through which firms gradually build productive capacity. Capital adjustment costs are a canonical friction in models of firm investment and plant dynamics ([Cooper and Haltiwanger, 2006](#)), and recent work shows that they can generate age-dependent responses to aggregate shocks ([Krusell et al., 2023](#)). In our model, young firms begin with low capital stocks and gradually adjust toward mature scale. Because capital is costly to adjust, young firms rely relatively more on flexible intermediate inputs. This makes input-cost reductions especially valuable early in the life cycle. The calibrated model reproduces the qualitative age pattern observed in the data: young firms respond more strongly to input tariff reductions, and the response weakens as firms mature. The model also predicts that input liberalization should induce stronger subsequent capital accumulation among young firms, a prediction we confirm in the data.

Our findings contribute to two different literatures. First, we contribute to the literature on heterogeneous firm responses to trade liberalization. Existing work has emphasized productivity, markups, input sourcing, exporting, innovation, and financial constraints as key determinants of firms' adjustment to trade shocks ([Melitz, 2003](#); [Manova, 2013](#); [Halpern et al., 2015](#); [Bloom et al., 2016](#)). We show that firm age is an additional and relevant dimension of heterogeneity. The age gradient survives flexible controls for the standard dimensions through which the trade literature usually measures firm hetero-

geneity. This suggests that age captures a distinct dynamic component of firms' exposure to input liberalization.

Second, we contribute to the firm dynamics literature, which documents that young firms grow faster, are more volatile, and differ systematically from mature firms in their behavior and performance (Evans, 1987; Hopenhayn, 1992; Haltiwanger et al., 2013). Our results show that these life-cycle differences also shape how firms respond to external policy shocks. Trade liberalization does not affect a representative firm; it affects firms at different stages of development, with different accumulated capabilities and different margins of adjustment. Therefore, the age composition of firms affects the transmission of external shocks. The present paper bridges these two streams of literature by examining how firms' positions in their life cycle influence their responses to trade liberalization.

The results have implications for how we evaluate trade reform. The absence of a differential age effect from output tariff reductions suggests that young firms are not disproportionately hurt by increased import competition. At the same time, the stronger response of young firms to input tariff reductions implies that trade liberalization can shift gains toward the early part of the firm life cycle. Because payoffs early in a firm's life are especially important for the value of entry, input liberalization may increase incentives to create new firms and may accelerate the growth of young producers. More broadly, our evidence suggests that firm age provides a simple but powerful way to understand heterogeneity in the effects of trade policy.

The rest of the paper is structured as follows. Section 2 discusses the data used and Section 3 provides more background to the trade liberalization episode in India. The baseline results are presented in Section 4. Section 5 shows empirical evidence against standard mechanisms that could be driving our results. Section 6 presents a model highlighting adjustment costs as a mechanism and quantifying its importance. Finally, Section 7 concludes.

2 Data

For our main results we use data on Indian firms from PROWESS. This includes panel data at the firm level and at the product level. Firm-level variables we use include year of incorporation, total sales revenue, capital, labor, materials, export earnings, financial frictions, and the industry in which the firm is (mainly) active. Age of the firm is defined as the number of years after incorporation.⁴

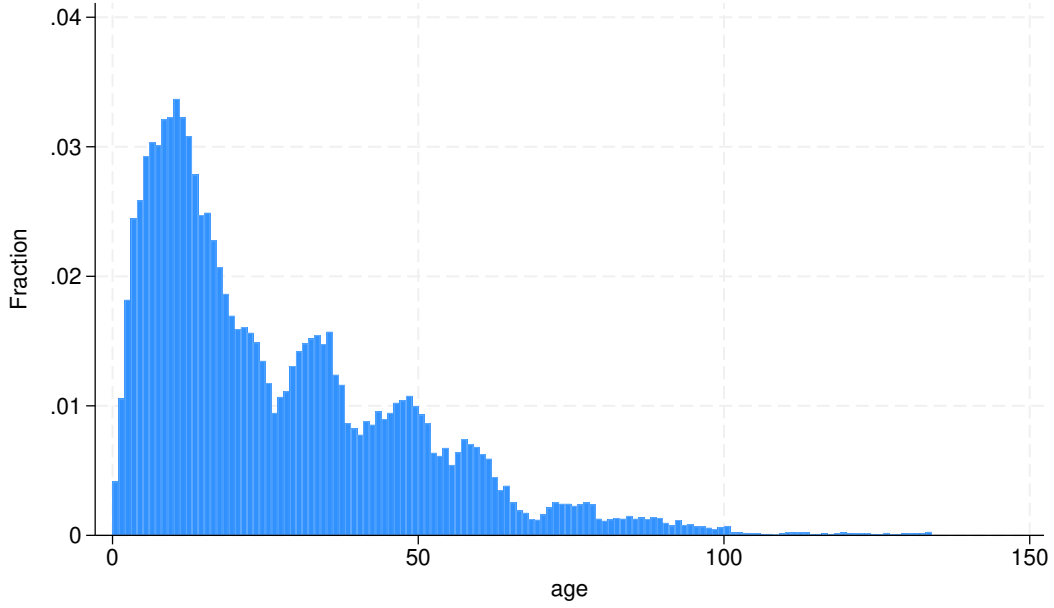
Product-level variables we use are sales quantity and its unit of measurement, and sales value. Products are classified according to 20-digit Centre for Monitoring Indian Economy (CMIE) codes, giving some 800 unique products in the estimation sample. In addition, we use data on Indian output and input import tariffs at the product level in the years 1989-1997 from [Topalova and Khandelwal \(2011\)](#) and [De Loecker et al. \(2016\)](#). The output tariffs are simply the observed import tariffs, while the input tariffs are constructed using the input-output table. Namely, the input tariffs are a weighted average of the output tariffs on all the inputs used to produce a given product, where the weights come from the 1993-1994 input-output table for India.

To reach our final estimation sample, we drop financial firms. We then merge in the product-level data, including the tariffs. In the tariff data, products are classified according to the National Industrial Classification (NIC), thus we make use of a concordance table in order to match these codes to the CMIE codes from the PROWESS data. We drop observations for which we do not have tariff data. Furthermore, we focus on the years 1989-1997 since [Topalova and Khandelwal \(2011\)](#) find some evidence against orthogonality of the tariff shock to sectoral productivity for the period after 1997. See [Appendix A](#) for more details on the data.

PROWESS does not cover the universe of firms. It contains information on all listed companies in India and a subset of unlisted companies. For our purposes, it matters whether firms of a given age are representative of all Indian manufacturing firms of that

⁴Note that the incorporation year is the year in which the company came into existence as a distinct legal entity, which does not always correspond to the year in which the enterprise came into existence. A company may have a relatively recent year of incorporation because of privatization, corporate restructuring, a de-merger, or because the enterprise existed as a partnership or as a proprietorship before it was incorporated as a company.

Figure 1: Histogram of firm age distribution



Note: The figure only includes firms that manufacture products for which tariff data is provided. Years 1989-1997.

age. [Figure 1](#) shows a histogram of the distribution by age. Although there are firms present for any age up to age 100, there are relatively few firms at the low end of the age distribution. If entry and exit rates by age are constant over time, we would have that the number of firms of a certain age in a given period is declining in age. This does not hold for the youngest firms as the number of firms by cohort is rapidly increasing until age 5. This suggests that selection is most worrisome for the youngest firms. Therefore, in our baseline analysis we exclude firms that are younger than six years old. [Figure 1](#) also shows that there are relatively many firms aged around 50. These are the firms that were founded around the time of India's independence.

We complement our analysis by using data from the Annual Survey of Industries (ASI), which provides information on registered manufacturing plants in India. ASI covers the universe of plants employing more than 50 workers and includes, in addition, a random one-third sample of plants with employment between 10 and 50 workers. We use years 1990-1993, 1994-1995, and 1997-1998 for which data is readily available. ASI is a repeated cross-section rather than a panel for the years we are interested in, and it does not report prices and quantities at the product level. Therefore, we do use PROWESS

as our baseline as it is a panel with detailed product-level data. However, ASI provides good coverage of firms at all ages, especially the very youngest. [Figure A.1](#) in [Appendix B](#) shows a smoother distribution of firms compared to [Figure 1](#) suggesting that ASI provides a more comprehensive coverage than PROWESS, especially for the youngest firms, and therefore we use it as a robustness check.

3 Background: trade liberalization in India

For several decades after independence, India pursued an import-substitution development strategy characterized by high tariffs, extensive non-tariff barriers, import licensing, and substantial state intervention in industrial activity. By the late 1980s, India was widely considered one of the most protected economies in the world. This policy regime changed sharply after the 1991 balance-of-payments crisis. That year, facing severe foreign exchange shortages, the government signed an agreement with the IMF, according to which India had to adopt a broad stabilization and structural reform program. Trade reform was a central component of this program and included large reductions in tariffs, the removal of many quantitative restrictions, and a simplification of the tariff schedule.

This episode has been exploited extensively in the trade literature as many of its characteristics conform a setting with a sufficiently exogenous shock to firms. First, the reduction of tariffs came as a surprise to firms as it was sudden, proposed by an external body and approved by a government where most figures were openly against these kinds of measures in the past ([Hasan et al., 2007](#)). The adjustment program was approved promptly and with little debate so as to avoid political opposition. Second, the aim of the tariff cut was to bring all tariffs to relatively similar and much lower levels. [Topalova \(2010\)](#) documents indeed a strong correlation between the pre-reform tariff levels and the decline in tariffs experienced by the industry. This resulted in the mean tariff falling from 128% in July 1991 to 35% in 1997, and the standard deviation falling from 41 percentage points to roughly 15 over the same period ([Hasan et al., 2012](#)). This harmonization of trade tariffs implies that the most affected sectors are the ones

that faced high tariffs initially, while trade policy was set earlier and remained largely the same as the economy evolved. Indeed, [Topalova and Khandelwal \(2011\)](#) show that output tariff changes during the trade liberalization were uncorrelated with preexisting firm or industry characteristics. [Goldberg et al. \(2010\)](#) present similar evidence for changes in the schedule of input tariffs. The literature has also argued that vested interests do not seem to have affected the intensity or timing of the tariff changes during this reform. In our setting, we indeed verify that there is no pre-trend in the sense that tariffs during 1991–1997 do not affect our outcome of interest during 1989–1990.

We follow this literature and focus on tariff changes over the period 1989–1997. This window captures the main liberalization episode that followed the 1991 crisis while avoiding later policy changes for which exogeneity is less clear. In particular, the literature has raised greater concerns about the post-1997 period, when tariff changes may have been more affected by industry lobbying or policy discretion. Restricting attention to the earlier reform period therefore helps isolate variation in trade policy that is more plausibly orthogonal to firm-level performance.

4 Results

In this section, we study how changes in India’s trade tariffs affect firm outcomes based on the age of the firm. The reform generated large declines in both output and input tariffs between the late 1980s and the late 1990s. As discussed in [Section 3](#), these reductions were broad-based and largely driven by the initial tariff structure: sectors with higher pre-reform tariffs experienced larger subsequent declines as the tariff schedule was compressed. This feature of the reform is useful for our purposes because it created substantial product-level variation in the change in trade protection, while the timing and intensity of the reductions were not chosen in response to contemporaneous firm performance.

We distinguish between two channels. Output tariffs measure protection in the markets in which firms sell their products. A decline in output tariffs increases import competition and should therefore put downward pressure on domestic firms’ sales, prices,

or quantities. Input tariffs measure protection on the goods used in production. A decline in input tariffs lowers the cost of intermediate inputs, either directly for firms that import those inputs or indirectly through lower domestic input prices. Firms face product-specific trade tariffs. This allows us to exploit variation in tariffs within a firm-product pair over time. That is, we control for firm-product fixed effects, as well as industry-year fixed effects. This gives the following regression equation,

$$y_{ijt} = \beta_1 \tau_{jt}^{out} + \beta_2 \tau_{jt}^{out} \times Age_{it} + \beta_3 \tau_{jt}^{in} + \beta_4 \tau_{jt}^{in} \times Age_{it} + \alpha_{ij} + \lambda_{st} + \varepsilon_{ijt}, \quad (1)$$

where y_{ijt} denotes the outcome variable for firm i , product j in year t . α_{ij} denotes the firm-product fixed effect, λ_{st} the industry-year fixed effect and ε_{ijt} the error term. Our coefficients of interest are β_2 and β_4 , which measure the extent to which older firms respond differently to changes in tariffs. We cluster standard errors at the industry level.

Note that equation (1) does not allow us to identify the effect of age by itself on the outcome variable. This is due to the presence of both firm fixed effects and time fixed effects, which absorb all variation in age. However, due to the variation in tariffs over time we do identify the differential response of tariff changes by age among firms in the same industry-year. Also, the presence of the firm fixed effect means that we control for cohort fixed effects as well. Equation (1) assumes that the differential effect by age is linear. Later in this section we discuss non-linear effects. Throughout the section, coefficients are reported as the effect of an increase in tariffs. Since the Indian reform mainly reduced tariffs, the effect of liberalization should be read with the opposite sign.

Table 1 shows the effect of the tariffs on log sales as reported in PROWESS. The first column ignores age and shows that higher output tariffs lead to higher sales. When output tariffs increase by 10 percentage points, log sales increase by 0.015. This coefficient is statistically significant. This result is expected as a higher import tariff on the product that a firm produces means that the firm is more protected from competition from abroad. The input tariff shows a negative coefficient, though imprecisely estimated. A negative effect is expected as a higher input tariff means it becomes more expensive to import the

Table 1: Effect of tariffs on log sales

	(1)	(2)
τ^{out}	0.150** (0.0583)	0.158 (0.136)
τ^{out} x age		-0.000505 (0.00440)
τ^{in}	-0.553 (0.559)	-1.389* (0.706)
τ^{in} x age		0.0417*** (0.0142)
Firm-Product FE	X	X
Industry-Year FE	X	X
N	25702	25702
R^2	0.118	0.120

Effects of an increase in the tariff by 100 percentage points. Firm and product-level data from PROWESS. This table only includes firms that are between 6 and 50 years old. Standard errors clustered at the industry level are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

inputs needed for production, therefore lowering the competitiveness of firms using those inputs.

The second column of [Table 1](#) introduces the interaction between tariffs and firm age. Compared to old firms, young firms do not respond differentially to changes in the output tariff. On the other hand, the interaction effect between age and the input tariff is positive and significant. When the input tariff increases by 10 percentage points a firm that is 10 years old lowers its sales by 0.1 log points, while a firm that is 30 years old does not, on average, change its sales after a change in input tariffs. [Table A.1](#) presents similar results when using ASI data, for which coverage of young firms is more complete.

Non-linear age effects

The linear specification in [Table 1](#) summarizes the average age gradient for firms between ages 6 and 50. We exclude firms younger than 6 in the baseline because PROWESS appears to underrepresent the youngest firms, and we exclude firms older than 50 be-

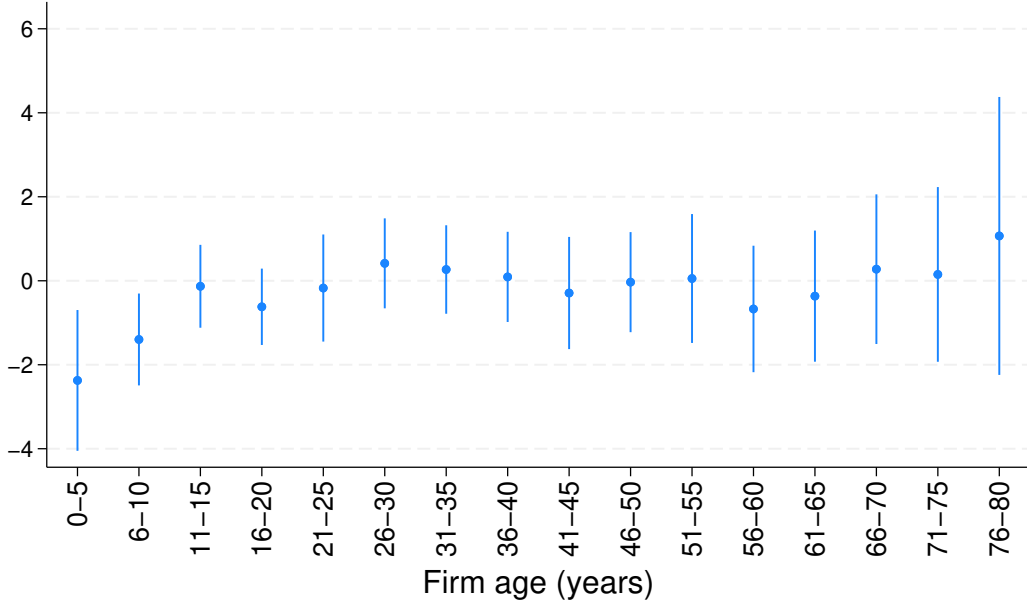
cause age differences among mature firms are less likely to capture meaningful life-cycle variation. However, the relationship between age and tariff responses need not be linear. A firm in its first decade may differ sharply from a firm in its second or third decade, while differences between mature firms may be much smaller.

To study the non-linear age effects, we group firms in age bins of five years,

$$y_{ijt} = \sum_b \beta_1^b \tau_{jt}^{out} \times Age_{it}^b + \sum_b \beta_2^b \tau_{jt}^{in} \times Age_{it}^b + \sum_b \beta_3^b Age_{it}^b + \alpha_{ij} + \lambda_{st} + \varepsilon_{ijt}, \quad (2)$$

where $Age_{it}^b = 1$ if firm i in year t is in age bin b and otherwise Age_{it}^b equals zero. Figure 2 reports the age-bin coefficients of the input tariff to graphically observe non-linearities in the results. For completeness, we also include here firms younger than six years old, and firms older than fifty.

Figure 2: Differential effect of input tariffs on log sales by firm age



Note: Point estimates and 95% confidence intervals for the effect of a change in tariffs on sales for firms of different age bins, as per equation (2). Firm and product-level data from PROWESS.

The figure documents a clear non-linear age relationship of the input tariff on sales. Young firms respond negatively in terms of sales with respect to an increased tariff while for older firms there is no such effect. The age threshold after which the effect disappears is not high: we find no effects for firms belonging to all age bins from 30 years old onward.

The differential age effect is significant on average, as shown by [Table 1](#), for firms between the ages of 6 and 50. Note that there is a non-monotonicity in the sense that firms aged 16-20 have a more negative response compared to both the firms directly younger and those directly older. Although this effect is not statistically significant, it is consistent with the model we discuss in [Section 6](#). Results for ASI, shown in [Figure A.2](#) in [Appendix B](#), give a very similar picture.

Using the point estimates, we can ask what fraction of the predicted sales gains from input tariff reductions accrue to firms below age 20. Given that we have year fixed effects, this is a partial equilibrium exercise. Our results imply that even though firms younger than 20 years account for 45% of sales in PROWESS, they account more than fully for the sales response to input liberalization (115%).

Consistent with our linear specification, [Figure A.3](#) in [Appendix C](#) shows that the effect of output tariffs is mostly homogeneous by age. The estimates fluctuate across age bins, but there is no systematic pattern.⁵

Prices versus quantities

The above effects on sales could be due to differential age effects on either prices or quantities, or both. This distinction matters for interpretation. If young firms respond to input liberalization mainly by lowering prices, the age gradient would reflect differential pass-through or markups. If they respond mainly by increasing quantities, the result is more consistent with cheaper inputs relaxing a production or expansion margin. [Table 2](#) shows the results of estimating equation (1) using log prices and log quantities as outcomes.

Recall that [Table 1](#) shows an overall positive effect of output tariffs on sales. [Table 2](#) highlights that this is due to a positive effect on prices while the output tariff has no effect on quantities (columns 1 and 3). This is consistent with output tariffs insulating domestic

⁵The exception is perhaps the effect for firms five years or younger. The coefficient is -0.5 and markedly different than the other coefficients. Panel (a) of [Figure A.2](#) in [Appendix B](#) presents results with ASI data where coverage of the youngest firms is less problematic and shows a similar picture. Confidence intervals for the youngest firms are smaller there and the effect for this group is slightly negative.

Table 2: Effect of tariffs on log prices and log quantity

	prices		quantities	
	(1)	(2)	(3)	(4)
τ^{out}	0.146** (0.0610)	0.190 (0.147)	-0.00258 (0.0775)	-0.0157 (0.178)
τ^{out} x age		-0.00175 (0.00490)		0.000295 (0.00566)
τ^{in}	0.447 (0.464)	0.674 (0.683)	-1.027** (0.415)	-2.119*** (0.566)
τ^{in} x age		-0.0116 (0.0146)		0.0546*** (0.0161)
Firm-Product FE	X	X	X	X
Industry-Year FE	X	X	X	X
N	24105	24105	24105	24105
R^2	0.113	0.113	0.038	0.042

Effects of an increase in the tariff by 100 percentage points. Firm and product-level data from PROWESS. This table only includes firms that are between 6 and 50 years old. Standard errors clustered at the industry level are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

producers from import competition. Moreover, the negative effect of input tariffs found above is associated with a negative effect on the quantity sold, compatible with cheaper inputs.

Focusing on the age-tariff interaction effect, [Table 1](#) does not show a differential effect of the output tariff on revenue by age. In theory, this could be because there are two opposing effects on prices and quantities. [Table 2](#) shows that this is not the case and that there is no differential effect by age of the output tariff on both prices and quantities. Finally, column 4 shows that the positive age effect of input tariffs on revenues is due to a positive effect on quantities, while there is a weak negative effect on prices (column 2). Thus, when input tariffs decline, young firms expand quantities more than old firms, but do not (strongly) differentially reduce prices. This pattern suggests that input liberalization affects young firms primarily by allowing them to produce and sell more, rather than by inducing stronger price pass-through.

How is it possible that quantities respond to changes in input tariffs while prices do not respond? One interpretation is that markets are competitive such that firms sell at their marginal cost while facing decreasing returns to scale. A decline in input tariffs lowers marginal costs, to which firms respond by increasing their production until their marginal cost is again equal to their original marginal cost. Thus, this generates no change in prices but a change in quantities.

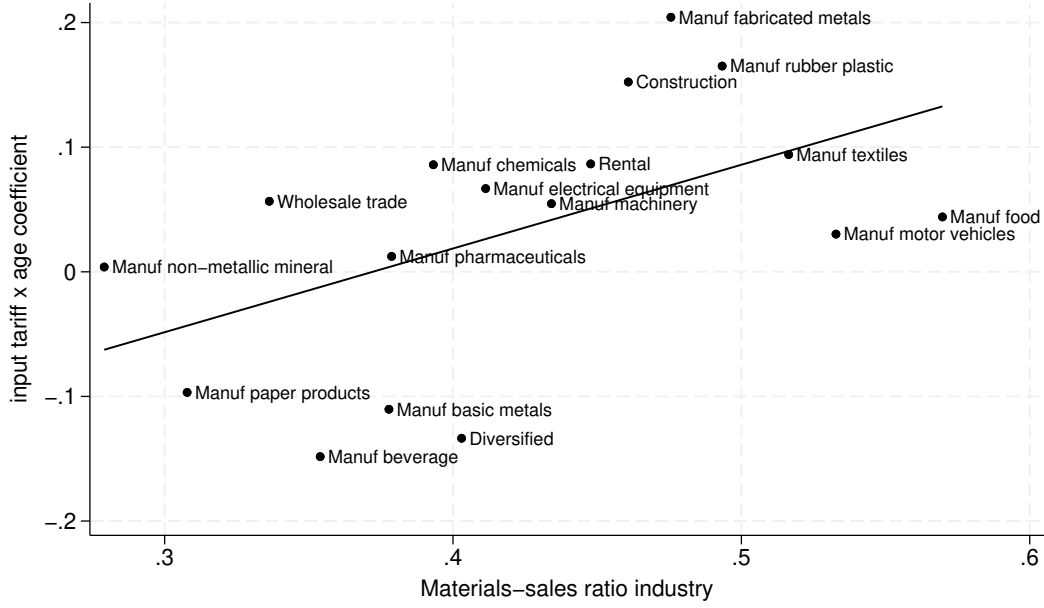
Also here, the effect on prices and quantities could be non-linear. [Figure A.4](#) shows there is some variation by age in terms of the effect of tariffs on prices, but no clear pattern emerges. This is in line with the results of [Table 2](#). [Figure A.5](#) shows there are no strong non-linear age effects of output tariffs on the quantity (except for the youngest age category). However, the second panel of this figure shows that firms up to 20 years are more negatively affected by input tariffs in terms of quantity sold than older firms. Overall, these results indicate that quantities and not prices are affected by input tariffs and that the effect is driven by younger firms.

Across industry variation

Finally, we examine whether the age gradient is stronger in industries where intermediate inputs are more important. If the mechanism operates through input-cost reductions, then the interaction between input tariffs and age should be larger in industries with higher materials intensity.

For each industry with at least 500 observations, we estimate equation (1) separately and recover the coefficient on the interaction between input tariffs and age. [Figure 3](#) plots these coefficients against the industry's materials-to-sales ratio. The age gradient varies substantially across industries, but it is systematically larger in more materials-intensive industries. The effect is strongest in the manufacturing of fabricated metals industry, with a coefficient of 0.24. The effect is smallest in the manufacturing of beverages industry. The fitted line has a positive slope of 0.7 and is statistically significant at the 5 percent level. This cross-industry pattern supports the interpretation that the differential response of

Figure 3: Input tariff x age coefficient at the industry level versus materials intensity



Note: The vertical axis reports the point estimates for the interaction input tariffs x age, estimated as in equation (1) per industry. The horizontal axis measures the materials to sales ratio of the industry. Firm and product-level data from PROWESS.

young firms is tied to input-cost exposure rather than to a generic age-specific sensitivity to policy changes.

5 Mechanisms

The results above show that young firms respond more strongly to input tariff changes than older firms. This section asks what accounts for that age gradient. We first examine whether the differential response is absorbed by observable firm characteristics that vary systematically over the life cycle, such as size, productivity, product scope, markups, input mix, trade exposure, financial position, and product quality. We then ask whether the age gradient reflects selection rather than within-firm life-cycle differences. Finally, we examine firm-level outcomes to identify which margins of adjustment move after input tariff changes.

To do so, we augment the baseline specification with observable firm characteristics and interact them with tariffs,

$$y_{ijt} = \beta_1 \tau_{jt}^{out} + \beta_2 \tau_{jt}^{out} \times Age_{it} + \beta_3 \tau_{jt}^{in} + \beta_4 \tau_{jt}^{in} \times Age_{it} + \beta_5 x_{it} + \beta_6 \tau_{jt}^{out} \times x_{it} + \beta_7 \tau_{jt}^{in} \times x_{it} + \alpha_{ij} + \lambda_{st} + \varepsilon_{ijt} . \quad (3)$$

The observable firm characteristics are denoted by x_{it} , which vary only at the firm level as does age.

This exercise should not be interpreted as a causal decomposition of the age effect. Several of the characteristics we consider in x_{it} may themselves evolve in response to tariff changes, and some are jointly determined with firm performance. The purpose of the exercise is instead diagnostic: if the age interaction were primarily capturing heterogeneity by size, productivity, import intensity, or another observable characteristic, controlling for that characteristic and its interaction with tariffs should substantially attenuate the age coefficient. Suppose that the effect of age is completely explained by large firms responding differently to tariffs compared to small firms, and that old firms differ in their size compared with young firms. In that case, if x_{it} denotes firm size, we would expect that β_2 and β_4 become zero when x_{it} is included. On the other hand, when adding the variable x_{it} does not substantially change the coefficients β_2 and β_4 compared to our baseline estimate, this suggests that the age effect does not come from differences in x_{it} along the life cycle.

[Table 3](#) evaluates whether the age gradient is explained by standard observable dimensions of firm heterogeneity. It shows the effects on the tariff times age interaction term when a range of observable firm characteristics are included one by one. We aim to cover a broad set of observable firm characteristics and include variables that cover different relevant dimensions of the firm: their size, efficiency level, market power, input mix, product scope, financial position, openness level, and the quality of the products they sell. If age simply proxies for some of these observable characteristics, controlling for that characteristic and its interaction with tariffs should substantially attenuate the

Table 3: Effect of tariffs on log sales with mediation variables

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
τ^{in}	-1.389*	-0.217	-1.522**	-1.450**	0.161	-0.854	-0.997	-1.285*	-1.685**	-1.841*	1.171	-1.461*	-1.613**
	(0.706)	(0.997)	(0.728)	(0.732)	(1.510)	(0.893)	(1.184)	(0.705)	(0.785)	(1.043)	(2.024)	(0.777)	(0.769)
$\tau^{in} \times \text{age}$	0.0417***	0.0357**	0.0420***	0.0422***	0.0449**	0.0460***	0.0301*	0.0374***	0.0523**	0.0454***	0.0379	0.0393***	0.0458***
	(0.0142)	(0.0160)	(0.0151)	(0.0152)	(0.0187)	(0.0148)	(0.0172)	(0.0140)	(0.0211)	(0.0167)	(0.0464)	(0.0146)	(0.0140)
x		0.173***	0.0101	-0.00555***	0.884***	-0.114	-0.0821*	0.484***	0.362**	0.00187	-5.950***	-0.628***	0.255***
		(0.0322)	(0.00821)	(0.00164)	(0.0496)	(0.0689)	(0.0469)	(0.0435)	(0.157)	(0.00469)	(0.898)	(0.0516)	(0.0745)
$\tau^{in} \times x$		-0.156	0.0309	0.00616	-0.512*	-0.898***	-0.0664	0.00605	-1.001	-0.00512	8.302	0.165	-0.407
		(0.102)	(0.0468)	(0.00488)	(0.289)	(0.319)	(0.216)	(0.180)	(1.602)	(0.0170)	(5.539)	(0.281)	(0.398)
Mediation var	None	sales (log)	# products	TFPQ (log)	sales/L (log)	markup (log)	K/L (log)	M/K (log)	export/sales	import M (%)	fin fric	liab/sales (log)	quality
Firm-Product FE	X	X	X	X	X	X	X	X	X	X	X	X	X
Industry-Year FE	X	X	X	X	X	X	X	X	X	X	X	X	X
N	25702	22231	25469	18317	25262	25213	25218	25101	16455	10564	10223	25287	23635
R^2	0.120	0.125	0.121	0.155	0.194	0.132	0.132	0.192	0.150	0.119	0.064	0.183	0.129

Effects of an increase in the tariff by 100 percentage points. Firm and product-level data from PROWESS. This table only includes firms that are between 6 and 50 years old. Standard errors clustered at the industry level are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

age coefficient. Since the output-tariff interaction is not statistically significant in the baseline, we only include the coefficients on the input tariff in [Table 3](#). However, the regression does also include the output tariff as shown in equation (3).

The first column repeats the baseline result from [Table 1](#) showing that there is a positive interaction effect of input tariffs and age. The second column shows that when x_{it} is equal to firm size measured by firm-level sales (in logs), the larger firms have higher product-level sales (0.173), and that there is a negative interaction effect between firm size and the input tariff (-0.156) though insignificant. However, the interaction effect between age and the input tariff, our coefficient of interest, changes only little when firm size is included in the regression. This implies that the age interaction effect is not directly due to firms differing in their size by age.

It is relevant to note here that as the left-hand side is product-level sales there is by construction a relationship between x and y . For instance, for single-product firms all variation in y would be absorbed by x in this case. Likewise, also for multi-product firms there is a mechanical relationship as high sales at the product level will also lead to high sales at the firm level. Therefore, to measure the effect of firm size we measure firm size here as firm-level sales of the other products each firm is producing (thus, this equals firm sales leaving out sales of the product that is on the left-hand side). We only run this regression for multi-product firms. Column 3 uses the number of products each firm is producing as an alternative measure of firm size. This alternative size measure also leaves the age-tariff coefficient essentially unchanged.

The fourth column includes productivity as a relevant characteristic. We take as measure of productivity TFPQ as estimated by [De Loecker et al. \(2016\)](#). The interaction effect between age and the input tariff, our coefficient of interest, is virtually unchanged when TFPQ is included in the regression.⁶ Productivity can also be measured in other ways than total factor productivity. Column 5 shows that also when productivity is measured by sales per worker our results are unchanged.

⁶Also using TFPR at the firm level as a characteristic does not change the interaction effect between age and the input tariff.

Markups provide another possible explanation for heterogeneous responses. We measure markups by multiplying the output elasticity with respect to materials with sales divided by material expenditure. We assume a Cobb-Douglas production function within industry such that when we take logs the output elasticity is absorbed by the industry times year fixed effect. Column 6 shows that firms' responses to tariff changes vary with markups, but controlling for this margin does not attenuate the age coefficient.

Next, we study whether differences in capital-labor intensity can explain the age effect. Column 7 shows that the age effect is indeed muted when the capital-labor intensity is included. The coefficient of interest goes from 0.0417 to 0.0301. This attenuation suggests that differences in quasi-fixed input intensity may account for part of the age gradient, but they do not fully explain it. Column 8 shows that including the material intensity of production, measured by the materials-capital ratio, leads to a slightly weaker age effect.

Columns 9 and 10 control for the trade intensity of firms, both in terms of exports and imports of materials. In both cases the effect on the age coefficient is modest. The modest role of imported-input intensity is informative. It suggests that the input-tariff channel is not limited to firms that directly import a large share of their materials. This is consistent with input tariff reductions also lowering domestic input prices through competitive pressure on domestic suppliers, as documented in [Table 2](#).

One other reason for why firms might respond differently by age is that young firms are more financially constrained than old firms. Therefore, column 11 controls for a measure of financial frictions as proposed by [Whited and Wu \(2006\)](#). The point estimate of the age coefficient does not change much compared to the baseline, but the coefficient is now insignificant due to an increased standard error. The increased standard error results from a decline in the number of observations due to not all firms reporting on each variable that is needed to construct the financial frictions indicator. Therefore, as an alternative we also measure financial frictions by liabilities over sales. The advantage of this measure is that we observe it for almost all firms. Column 12 shows that this leads to a similar age coefficient as in column 11 but now with smaller standard errors and therefore the coefficient remains statistically significant.

Next, we study whether differences in quality of the goods produced across firms can explain our result. We recover the quality of goods produced by the firm following [Khandelwal \(2010\)](#), as a measure of the size of the market share of a product-firm conditional on the price charged. Column 13 shows that, once more, this does not lead to a large change in the age coefficient.

The main takeaway from [Table 3](#) is that the age gradient is not absorbed by any of the observable characteristics considered here for x . Thus, the differential response by age can for instance not be attributed to differences in size or productivity across age groups. The firm characteristic that has the largest effect is the capital-labor ratio. But after taking into account differences in capital intensity, the age input tariff coefficient is still 0.0301.⁷

Another possibility is that the product-level sales response may reflect reallocation within the firm rather than an expansion of firm activity. If a tariff decline for one input leads a firm to shift inputs or managerial attention toward the corresponding product, sales of other products may fall. In that case, the product-level age gradient would overstate the firm-level response. However, we do not find any evidence for such substitution. [Table 4](#) shows that tariffs on a particular input do not affect the sales of the other products the firm is producing. Because tariffs are likely to be correlated for the different products that a firm produces, we control here for the sales-weighted average tariff among the other products each firm produces to capture the substitution effect. Tariffs on a given product do not significantly affect sales of the firm's other products once we control for the sales-weighted average tariffs faced by those other products.

Age effect: Selection vs. fundamentals

The previous exercise shows that the age gradient is not absorbed by observed firm characteristics. We next ask whether the gradient reflects changes that occur as firms age, or instead selection across cohorts. If firms with low responsiveness are more likely

⁷Following a similar procedure, we consider how adding different characteristics affects the interaction term between age and input tariffs when the outcome variable is product-level quantity. Again, the observable characteristic does not remove the age-input tariff interaction coefficient.

Table 4: Effect of tariffs on log sales of other products firms produce

	(1)
	firm-level sales other products
τ^{out}	0.0911 (0.125)
τ^{out} x age	-0.00137 (0.00415)
τ^{in}	-0.231 (1.197)
τ^{in} x age	-0.00219 (0.0379)
Firm-Product FE	X
Industry-Year FE	X
N	18655
R^2	0.344

Effects of an increase in the tariff on product x by 100 percentage points on the sales of the firm that are not product x . Firm and product-level data from PROWESS. We control for the sales-weighted average tariffs on the other products the firm is producing. This table only includes firms that are between 6 and 50 years old. Standard errors clustered at the industry level are reported in parentheses.

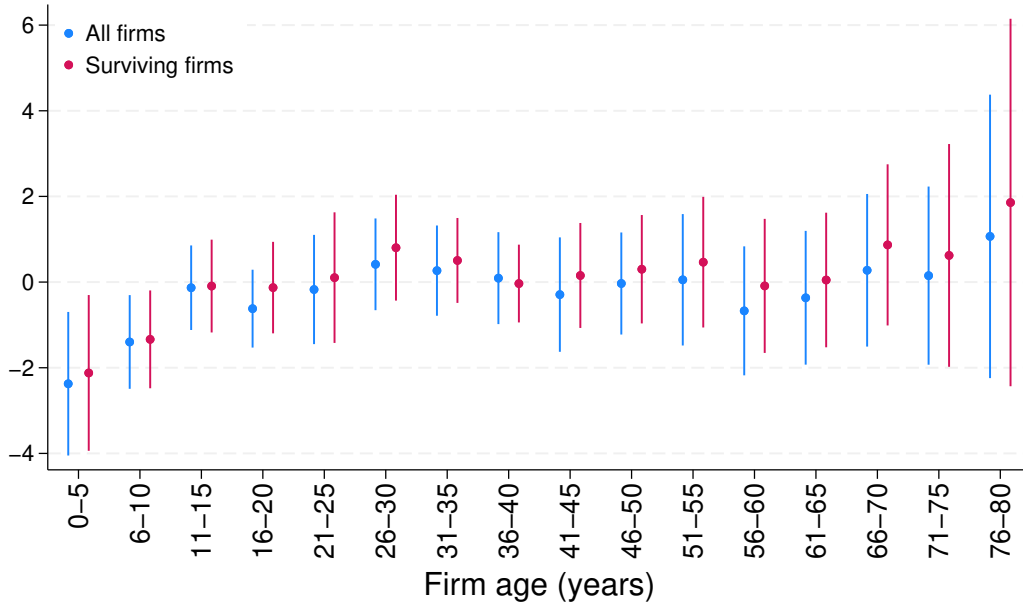
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

to survive, older cohorts may appear less responsive purely due to selection, even if individual firms' responsiveness does not change with age. In a long panel this would be picked up by the firm fixed effects, but we observe each firm over a limited part of their life cycle only.

In order to check whether our results are driven by selection, or rather, whether age affects firm fundamental characteristics, we compare firms within age groups conditional on survival. Although we consider the trade shock only until 1997, our data continue beyond that year. This allows us to condition on survival for more than 20 years after the trade shock. If firms that survive for an additional 20 years respond similarly to the trade shock as the full set of firms that are present during the trade liberalization, then it suggests that different reactions across age groups to the trade shock are in fact due to the age difference and not due to selection. On the other hand, selection would imply that the effect among all firms of a certain age is different from the effect among the subset

of firms of that age which survive at least 20 more years. For example, consider firms of age 10 and of age 30 during the trade reform and suppose they respond differently. Then, consider the subset of firms of age 10 that survive until at least age 30. If the difference between the 10 and 30-year-old firms is due to an age effect, then the subset of 10-year-old firms surviving until 30 should respond similarly to the full set of 10 year old firms. Conversely, if the difference is due to selection, this subset of firms should instead respond similarly to the 30-year-old firms.

Figure 4: Effect of input tariffs on log sales for all firms and those that survive at least 20 more years



Note: Point estimates and 95% confidence intervals for the effect of a change in tariffs on sales for firms of different age bins, as per equation (2). Results in blue include all firms belonging to the specific bin. Results in red are obtained for the subset of firms that survive for 20 years or more. Firm and product-level data from PROWESS.

Figure 4 shows the results of such an exercise in which we compare, within each age bin, the full set of firms to those that survive for at least twenty more years. The effect of input tariffs on sales by age looks very similar comparing the baseline and conditioning on survival for another twenty years. Although the point estimates are weakly higher, they are well within the confidence intervals for the full sample.⁸ We interpret this as evidence that selection is unlikely to be the main driver of the age gradient.

⁸When we consider firms with a survival horizon of 10 years, the results are even closer to the full sample.

The identifying assumption in this test is that the selection mechanisms firms faced to reach a certain age in the 1990s are the same as the selection mechanisms firms faced afterwards. Or, at least the firms selected in either period should have the same reaction to trade shocks. This would be violated if the trade shock itself may have changed the survival pressures. However, this would likely lead to a downward bias in the coefficients of the surviving firms. This is because the tariffs were reduced and therefore firms with lower coefficients had better outcomes following the shock. Yet, the coefficients on the surviving firms are not consistently below that of their non-surviving counterparts.

Effects on firm-level outcomes

Our baseline result was identified using variation in tariffs at the product level. This allows us to study how sales, prices and quantities at the product level respond. It is also interesting to see how the tariffs affect young and old firms differently in terms of, for instance, their productivity. However, as we observe inputs only at the firm level rather than at the product level, we can only do this at the firm level. Therefore, we construct the sales-weighted average input and output tariff the firm faces across products in each year.⁹

As the identification for these additional results depends on firm-level variation while the results so far use product-level variation within a firm, we first verify whether our main result on sales also holds using variation at the firm level, where we include firm fixed effects and year fixed effects. The first column of [Table 5](#) shows this is indeed the case. As in [Table 1](#), there is no differential age effect of the output tariff on firm-level sales, but there is a positive differential age effect of the input tariff. Also, the magnitude of the effect is very similar to the baseline.

The second column shows that the tariffs have no effect on total factor productivity and that there is also no differential effect of age. It is thus not the case that young firms

⁹One issue in doing so is that when tariffs change, firms change their product mix. In theory, it is therefore possible that if the tariff for a product declines the sales-weighted average increases. This would be the case if the tariff decline occurs for a product with a high initial tariff. The decline in the product-level tariff induces the firm to substitute towards the high-tariff product which could increase the sales-weighted average if this substitution effect is large enough. To avoid this, we hold the weights constant over the years and use as weights the sum of sales over the entire time period.

Table 5: Effect of tariffs on firm-level outcome variables

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
	sales (log)	TFPQ (log)	L-prod (log)	import M (%)	K/L (log)	M/K (log)	# goods (log)
τ^{out}	0.101 (0.125)	-2.450 (3.850)	-0.0319 (0.0766)	1.764 (1.084)	-0.194* (0.0997)	0.153 (0.117)	0.0408 (0.0798)
τ^{out} x age	0.000371 (0.00393)	0.0541 (0.137)	0.00170 (0.00239)	-0.0433 (0.0335)	0.00548* (0.00316)	-0.00823** (0.00385)	0.000153 (0.00274)
τ^{in}	-1.401*** (0.415)	5.220 (14.72)	0.606** (0.247)	-9.229** (4.472)	0.258 (0.350)	-0.460 (0.411)	-0.0780 (0.285)
τ^{in} x age	0.0520*** (0.0119)	-0.263 (0.422)	-0.0264*** (0.00735)	0.0365 (0.168)	-0.0213** (0.0101)	0.0203* (0.0121)	0.0191** (0.00843)
Firm FE	X	X	X	X	X	X	X
Year FE	X	X	X	X	X	X	X
N	13533	10660	13497	6526	13503	13396	13674
R^2	0.378	0.010	0.078	0.007	0.014	0.080	0.059

Effects of an increase in the tariff by 100 percentage points. Firm-level tariffs are weighted averages of the tariffs on all products the firm produces. Firm and product-level data from PROWESS. This table only includes firms that are between 6 and 50 years old. Robust standard errors are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

respond more strongly to input tariffs because they increase their productivity after a decline in tariffs. The third column, however, shows there is a negative interaction effect between age and the input tariff on labor productivity, where labor productivity equals firm-level sales divided by the wage bill. At first glance, this may seem surprising as our baseline result is a positive interaction effect on sales. However, this result does not imply that young firms respond more positively to a decline in input tariffs because their productivity declines relative to old firms. Rather, it is indicative of production featuring decreasing returns to scale. With decreasing returns to scale, labor productivity declines when firms expand in size while holding their TFPQ constant.

Furthermore, input tariffs have a negative effect on the imports of materials which is expected (column 4). There is no age gradient in this effect. Young firms thus do not seem to disproportionately benefit through the direct effect of improved access to foreign inputs. Rather, [Table 2](#) shows that when tariffs decline also prices of domestically sourced inputs decline. This means that firms may benefit from tariff reductions even if they do not import directly or change their import intensity.

Columns 5 and 6 examine input choices. There is a negative age-input tariff interaction effect on the capital-labor ratio. Thus, young firms that increase their sales after a decline

in input tariffs experience a decline in the capital-labor ratio. This is consistent with labor being a flexible input while capital is fixed within the period. Likewise, there is a positive differential effect on the materials-capital ratio. This suggests that labor and materials are both flexible inputs. Finally, column 7 shows that young firms do not only respond on the intensive margin, but also on the extensive margin. After a decline in input tariffs, young firms do increase the number of products they produce, and do so differently than old firms.

Taken together, the results of this section narrow the set of plausible mechanisms. The age gradient is not explained by standard observables such as size, productivity, markups, trade exposure, financial constraints, or quality. Nor does it appear to be driven primarily by selection or by reallocation across products within the firm. The one observable that meaningfully attenuates the age interaction is capital intensity, and the firm-level results show that young firms expand while their capital-labor ratio falls after input tariff reductions.

The empirical evidence therefore points to three features that any successful mechanism should account for. First, the age gradient appears only for input tariffs, not output tariffs. Second, it operates primarily through quantities rather than prices. Third, it survives controlling for standard observable characteristics but weakens once differences in capital intensity are taken into account. Together, these findings suggest that young firms adjust along flexible input margins while slower-moving productive inputs constrain their short-run scale. The next section develops a firm life-cycle model that captures this mechanism through adjustment costs in a quasi-fixed input.

6 Model

The empirical results point to firm age as a state variable that matters beyond size, productivity, product scope, markups, and import exposure. This section studies one mechanism that remains consistent with the evidence: sluggish adjustment of quasi-fixed factors. We show how this mechanism generates larger responses to input price changes

among young firms. This mechanism connects our evidence to a broader literature on firm life cycles, slow growth toward efficient scale, and frictions in capital accumulation. [Hsieh and Klenow \(2014\)](#) show that plants in India and Mexico grow much less over their life cycle than comparable U.S. plants, suggesting that barriers to scaling are quantitatively important in precisely this setting. [Akcigit et al. \(2021\)](#) emphasize that organizational frictions can limit firm growth in developing economies, while [Cooper and Haltiwanger \(2006\)](#) show that capital adjustment costs are central for understanding plant-level investment dynamics. Our model applies these insights to trade liberalization: if young firms are still adjusting toward their desired input mix, a reduction in input prices affects not only contemporaneous marginal costs but also the firm’s position along its growth path. Input liberalization can therefore have larger effects early in the life cycle, even among firms with similar current size or productivity.

In the model, we use physical capital as a tractable proxy for the broader set of quasi-fixed inputs that firms accumulate over time. Young firms begin operations below their frictionless scale and converge gradually because capital is costly to adjust. During this transition, they rely relatively more on flexible intermediate inputs. Input tariff reductions therefore relax a margin that is especially important early in the firm life cycle.

Besides capital, our model includes intermediate inputs as productive factors. This allows us to study the effect of a change in input prices. For notational simplicity and because there are no effects on the material-labor ratio, we do not include labor as a third factor of production. That is, we take production of firm i in year t as CES in capital (k) and materials (m):

$$y_{it} = z_{it} [\delta k_{it}^\rho + (1 - \delta)m_{it}^\rho]^{\frac{\gamma}{\rho}} \quad (4)$$

where $\delta \in (0, 1)$ denotes capital intensity, z represents total factor productivity, and $\rho \in (-\infty, 1)/\{0\}$ drives the elasticity of substitution, which equals $1/(1 - \rho)$. Production features decreasing returns to scale, $\gamma < 1$, which is consistent with the above evidence regarding the effect of tariffs on labor productivity. We assume markets are competitive, which is consistent with the effect of input tariffs operating through quantities and not

through prices. The output price is set as the numeraire. In the remainder we omit the firm and time subscripts for notational simplicity.

Firms can choose their materials freely, while they choose their capital input one period ahead and have to pay convex adjustment costs if their choice of capital differs from their current capital level. This leads to the following Bellman equation where $V(z, k, a)$ denotes the value of the firm as a function of its productivity, capital stock and age a :

$$V(z, k, a) = \max_{I, k', m} y(k, m) - p_m m - p^I I - \Psi(k, k') + \beta(1 - P_{\text{exit}})\mathbb{E}[V(z', k', a + 1)] \quad (5)$$

$$\text{s.t. } k' = (1 - \delta_k)k + I$$

$$\log(z') = \rho_a + \rho_z \log(z) + \varepsilon, \quad \varepsilon \stackrel{\text{iid}}{\sim} \mathcal{N}(0, \sigma_z^2). \quad (6)$$

Investment of the firm is denoted by I , k' denotes the capital stock the firm chooses for the next period and capital depreciates at the rate δ_k . The firm discounts at the rate β , and there is an exogenous exit rate denoted by P_{exit} . Productivity follows an AR(1) process, where we allow the intercept ρ_a to depend on firm age a . In the calibration, we parametrize ρ_a such that average productivity grows exponentially during the first years the firm operates. This is necessary to match the growth of firms over the life cycle quantitatively. The dependence of the productivity process on firm age makes that age becomes a state variable.

The price of materials is denoted by p_m , and the effect of the input tariff will operate through a change in this input price. This is a simplification as not all firms import their intermediate inputs. However, as shown in [Table 2](#), a decline in output tariffs lowers the domestic price of that product. As an example, suppose there is a car producer that buys steel domestically. According to this result, a decline in the steel tariff also lowers the price of domestically produced steel through competition. Therefore, this car producer will also face lower input prices.

In this model, a life cycle occurs for two reasons. The first is that productivity depends on age as discussed above. The second reason comes from the capital-adjustment costs

$\Psi(k, k')$. Firms start with a low capital stock, $k_1 > 0$, and due to the convex adjustment costs it is optimal to only slowly increase the capital stock over time to the optimal level absent adjustment costs. The adjustment cost function looks as follows:

$$\Psi(k', k) = \frac{\chi_1}{2} \left(\frac{k' - k}{k} \right)^2 k, \quad \text{with } \chi_1 \geq 0. \quad (7)$$

The convex cost means that if the firm decides to change its capital stock it will do so without too large changes.

The maximization problem can be split in two stages. Conditional on a chosen capital stock, the firm statically chooses its materials. Then, taking this first choice into account, the firm dynamically chooses capital. Focusing on the static optimization problem of choosing materials first yields the following first-order condition

$$z\gamma(1 - \delta)m^{\rho-1} [\delta k^\rho + (1 - \delta)m^\rho]^{\frac{\gamma-\rho}{\rho}} = p_m. \quad (8)$$

Using this equation, we can study what happens to output when the price of materials declines all of a sudden. First, substitute out materials from the production function using the first-order condition to get the following implicit equation for output:

$$y = z \left[\delta k^\rho + (1 - \delta) \left(\frac{(1 - \delta)\gamma}{p_m} \right)^{\frac{\rho}{1-\rho}} z^{\frac{\rho}{(1-\rho)\gamma}} y^{\frac{(\gamma-1)\rho}{\gamma(1-\rho)}} \right]^{\frac{\gamma}{\rho}}. \quad (9)$$

Take the derivative on both sides with respect to p_m and rewrite to get the following expression for $\partial y / \partial p_m$ ¹⁰:

$$\frac{\partial y}{\partial p_m} = \frac{-\frac{\gamma}{\rho}(1 - \delta) \left(\frac{(1-\delta)\gamma}{p_m} \right)^{\frac{\rho}{1-\rho}} z^{\frac{\rho}{(1-\rho)\gamma}} y^{\frac{(\gamma-1)\rho}{\gamma(1-\rho)}} \frac{(\gamma-\rho)\rho}{\gamma(1-\rho)}}{1 - \frac{\gamma}{\rho}(1 - \delta) \left(\frac{(1-\delta)\gamma}{p_m} \right)^{\frac{\rho}{1-\rho}} z^{\frac{\rho}{(1-\rho)\gamma}} y^{\frac{(\gamma-1)\rho}{\gamma(1-\rho)}} \frac{(\gamma-\rho)\rho}{\gamma(1-\rho)}} \frac{\gamma}{\gamma - \rho} \frac{y}{p_m}. \quad (10)$$

¹⁰First take the derivative of (9) on both sides, and then bring all $\partial y / \partial p_m$ to one side. Simplify by using equation (9).

In the empirical section we study $\frac{\partial \log(y)}{\partial p_m}$ which is equal to $\frac{\partial y / \partial p_m}{y}$. Using the above equation, this is equal to

$$\frac{\partial \log(y)}{\partial p_m} = \frac{-B(k)}{1 - B(k)} \frac{\gamma}{\gamma - \rho} \frac{1}{p_m} \quad (11)$$

where we use the notation $B(k) = \frac{\gamma}{\rho}(1 - \delta) \left(\frac{(1 - \delta)\gamma}{p_m} \right)^{\frac{\rho}{1 - \rho}} z^{\frac{\rho}{(1 - \rho)\gamma}} y^{\frac{(\gamma - 1)\rho}{\gamma(1 - \rho)}} \frac{(\gamma - \rho)\rho}{\gamma(1 - \rho)}$. This function depends on the capital stock through output.

Our baseline result is that young firms respond more strongly to changes in the price of materials. In other words, our main finding is that $\frac{\partial^2 \log(y)}{\partial p_m \partial age} > 0$. In this model, in the absence of changes in productivity over time there is an increasing relationship between firm age and capital. Firms start with a low capital stock and due to the adjustment costs slowly increase their capital stock up until they reach a certain level at which they keep the capital stock constant. Therefore, in the special case of productivity being constant over the life cycle, the model is consistent with the empirics if $\frac{\partial^2 \log(y)}{\partial p_m \partial k} > 0$. Let's take the derivative of (11) with respect to capital to see under which conditions this cross-derivative is positive.

$$\frac{\partial^2 \log(y)}{\partial p_m \partial k} = \frac{-B'(k)}{(1 - B(k))^2} \frac{\gamma}{\gamma - \rho} \frac{1}{p_m} \quad (12)$$

$$= \frac{-1}{(1 - B(k))^2} (1 - \delta) \left(\frac{(1 - \delta)\gamma}{p_m} \right)^{\frac{\rho}{1 - \rho}} z^{\frac{\rho}{(1 - \rho)\gamma}} y^{\frac{(\gamma - 1)\rho}{\gamma(1 - \rho)} - 1} \frac{\partial y}{\partial k} \frac{1}{p_m} \frac{(\gamma - 1)\rho}{(1 - \rho)^2}. \quad (13)$$

The sign of this expression depends on ρ . When capital and materials are gross complements such that $\rho < 0$, this cross-derivative is negative.¹¹ And when capital and materials are gross substitutes such that $\rho > 0$, this cross-derivative is positive. Thus, the model is consistent with our empirical finding when capital and materials are substitutes. The reason is as follows. Young firms have a low capital stock and are relatively far away from the optimal capital stock absent any adjustment costs. When capital and materials are substitutes, young firms choose relatively high materials. Then, a change in the materials input price is going to affect these young firms with relatively materials-intensive

¹¹Recall that $\gamma < 1$ and that the marginal product of capital is positive. With constant returns to scale ($\gamma = 1$) and with Cobb-Douglas ($\rho = 0$) the cross-derivative is equal to zero.

production more. When capital and materials are complements, it is still the case that young firms have a relatively high materials-capital ratio, but the materials-output ratio is smaller for young firms.¹² Therefore, changes in the input price have a smaller effect on young firms' output in this case.

When productivity changes as firms age, there is an additional effect in the opposite direction. The increase in productivity means that the optimal capital stock absent adjustment costs increases with firm age. When productivity growth is high relative to the adjustment costs, optimal frictionless capital can grow faster than capital itself for some periods, which means that the older firms are further away from the frictionless optimal capital level and thus respond more strongly to changes in input prices. In the calibrated version of the model, we find that for most ages the first effect dominates such that younger firms respond more to the shock. But for some ages, the second effect actually dominates which gives a non-monotonic relationship in age of changes in input prices, which is consistent with our empirical findings.

In terms of modeling, it is important that the input factor k slowly adjusts while being gross substitutes for materials. In the data we only observe physical capital, and therefore we refer to k as physical capital, and that is also what we target in the calibration. But k can in principle also stand for other factors such as organizational capital. For instance, if firms get better over time in managing their inventory and reducing waste, young firms will be more materials-intensive, and the forces that make that young firms respond more to input tariffs are at play.

Calibration

The calibration of the model is designed to match two central features of firm life-cycle dynamics: growth toward mature scale and changes in input mix during that transition. The mature-firm materials-capital ratio pins down the long-run production mix; early capital growth pins down the initial distance from mature scale; the persistence of cap-

¹²The materials-output ratio, m/y , is proportional to $y^{\frac{\rho(1-\gamma)}{\gamma(\rho-1)}}$. This means that when $\rho > 0$, m/y is declining in y (and thus also in firm age), while it is increasing in y when $\rho < 0$.

ital relative to sales disciplines the strength of adjustment costs; and the decline in the materials-capital ratio over age identifies the substitutability between capital and materials. The productivity process then accounts for residual sales growth over the life cycle, rather than allowing adjustment costs alone to fit all growth moments.

Table 6 presents the parameter values selected. δ governs the capital intensity of production, and we set it to 0.11 to target the materials-capital ratio of firms between the ages of 30 and 50. k_1 denotes the starting capital stock which we set to target the change in the capital stock during the first 10 years. The capital-adjustment costs govern how fast capital evolves in response to a productivity shock. When adjustment costs are high firms will adjust capital less and capital becomes more persistent. It is indeed the case that when estimating an AR(1) on log capital and log sales, the autocorrelation of capital is higher than the autocorrelation of sales. We set χ_1 and the autocorrelation of the productivity process ρ_z to target these autocorrelations. The parameter governing the elasticity of substitution between capital and materials affects how the materials-capital ratio changes by age. Young firms are further away from the optimal capital stock absent capital-adjustment costs. They will therefore substitute toward materials and young firms have thus a higher materials-output ratio compared to old firms. This is also the case in the data. In the model, the more substitutable capital and materials are, the larger the change in the materials-capital ratio as firms age. Targeting the ratio of the materials-capital ratio for firms aged 5-10 to that for firms aged 30-50 gives $\rho = 0.6$. The returns to scale, γ , are externally set to 0.95, the depreciation rate, δ_k , to 0.06 and the discount factor, β , to 0.96, while the price of materials is normalized to 1. The exit probability is obtained from Chatterjee et al. (2025), who find it to be 3.5% for India. The productivity process is parametrized such that average log productivity in year 1 is equal to zero, and that productivity grows exponentially for the first 20 years by γ_z which is set to 0.35% per year. After age 20, average productivity stops growing. An age of 20 is chosen because in the data most of firm growth occurs during the first 20 years. The standard deviation of the productivity shocks is set to 0.0185 to match the standard deviation of sales among firms between the ages of 30 and 50.

Table 6: Calibration

	Description	Value	Target moment	Data	Model
δ	Capital intensity	0.11	m/k at old age	1.46	1.46
k_1	Capital at age 1	0.0169	$\log(k_{10}/k_1)$	0.70	0.73
χ_1	Capital-adjustment costs	2.5	Autocorrelation capital	0.996	1.000
ρ	Substitution between k and m	0.6	m/k at young vs old age	1.24	1.22
γ	Returns to scale	0.95	External		
p_m	Price of materials	1	Normalization		
δ_k	Depreciation rate capital	0.06	External		
β	Discount factor	0.96	External		
P_{exit}	Exit probability	0.035	Chatterjee et al. (2025)		
γ_z	Exponential productivity growth	0.0035	$\log(y_{20}/y_1)$	1.24	1.25
ρ_z	Autocorrelation productivity	0.97	Autocorrelation sales	0.989	0.988
σ_z	Std dev productivity shocks	0.0185	Var log sales at old age	1.55	1.57

Notes: Old age refers to firms between the ages of 30 and 50.

Figure 5: Life cycle according to model and data.

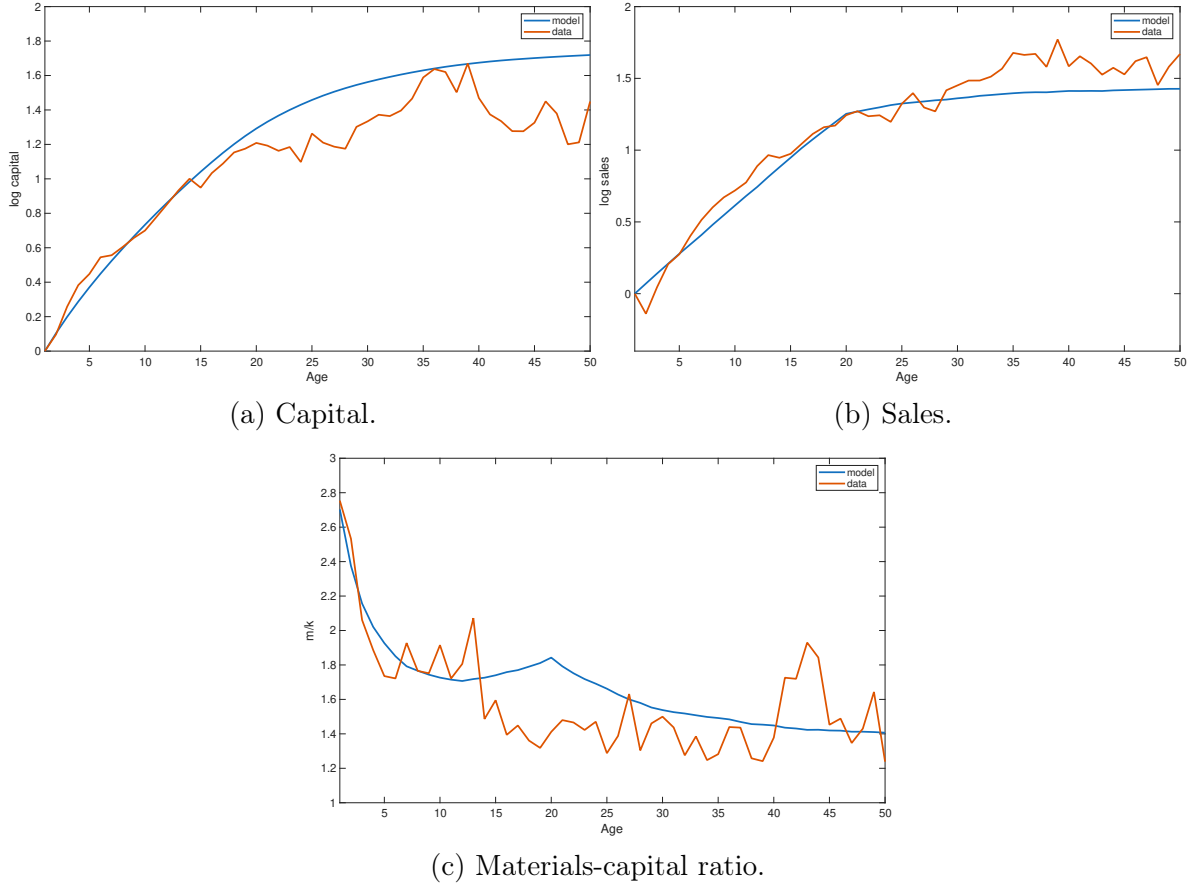


Figure 5a shows how capital evolves by age in both the model and data. The log of capital in the first year is normalized to zero. The model tracks the evolution of capital closely during the first 15 years, while for the later years the model overestimates the capital stock compared to the data. For sales the opposite pattern occurs (Figure 5b).

Also here, during the first years the model tracks the data closely, but after age 30, the model gives smaller sales compared to the data. [Figure 5c](#) shows that the model tracks the evolution of the materials-capital ratio over age well. We target the materials-capital ratio both at the ages 30-50 and the ages 5-10. But the model is also capable of capturing the rapid decline in the materials-capital ratio during the first 5 years.

Overall, [Figure 5](#) shows that the model reproduces the steep early accumulation of capital and the sharp decline in materials intensity that characterize young firms in the data. This is important for the mechanism: the model's tariff elasticity is not driven by age per se, but by the changing distance between actual capital and the frictionless capital stock.

Model results

Armed with the calibrated model, we can now replicate the data experiment in the model. To calculate the effect of a tariff change on sales in the model we do the following. Between 1989 and 1997 the average input tariff in our sample declined by 24 percentage points from 35% to 11%. Holding gross foreign input prices constant, this means that the price of imported intermediate inputs declines by 18% during this time period. By [Table 2](#), a decline in the (output) tariff of 100 percentage points leads to a decline in prices of 0.145 log points. Thus, the 24-percentage-point decline in input tariffs implies that domestically sourced inputs become 3.5% cheaper. On average, firms in our sample import 6% of their intermediate inputs. Hence, the price of the average input declined by 4.35% due to the tariff reduction. To run experiments in our model, we apply this 4.35% decline in input prices. And when we report the effects we divide the results by 0.24 to mimic the effect of a 100 percentage point decline in tariffs. [Figure 6a](#) shows the effect of a *positive* tariff change on log sales by age.¹³ Consistent with our main empirical finding, we find that younger firms respond more negatively to (positive) changes in input tariffs compared to old firms.

¹³The effect of a positive tariff change is taken as the negative of the effect of a tariff reduction. This mimics how we treat it in the data.

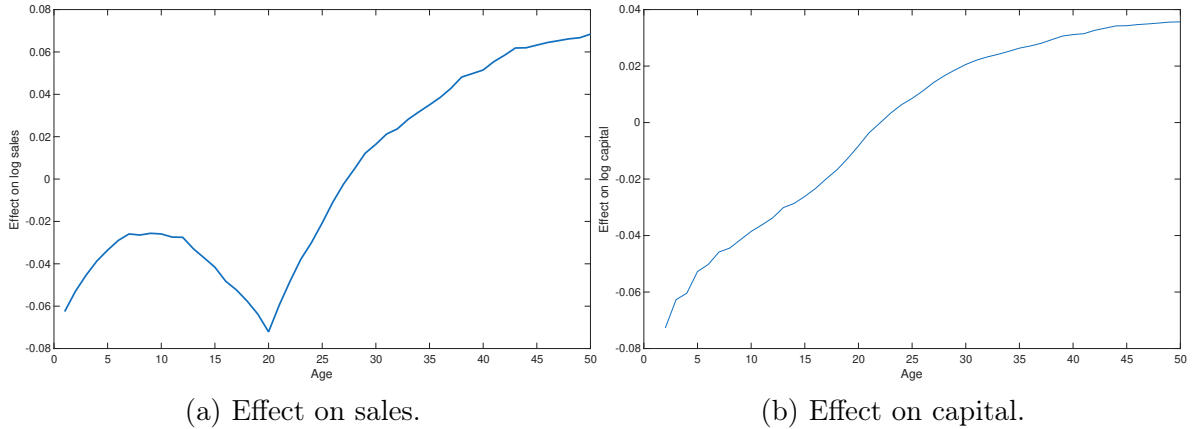
Interestingly, the model also replicates the non-monotonic behavior we found in [Figure 2](#). In both model and data, we find that firms around age 20 experience a more negative effect compared to the firms directly younger and older than them. In the model, this follows from how the productivity process is specified. Young firms start with a low capital stock such that they are far away from the capital stock if there were no adjustment costs. This implies that young firms respond strongly to the shock. In the years after, the capital stock grows relatively fast such that the effect of tariffs weakens. Between age 10 and age 20 the capital stock is still growing, but only modestly compared to the optimal capital stock absent adjustment costs which is growing due to the productivity growth. This increases the gap between actual capital and capital absent adjustment costs, and this increased gap means that the effect of tariffs on sales is becoming more negative again. After age 20, productivity stops growing while the capital stock continues to grow. Therefore, the effect of the tariff becomes more positive.

The pattern in [Figure 6a](#) is for an important part driven by productivity growth. If there were no or little productivity growth, the effect of tariffs would be monotonically increasing in age. If instead productivity grew (substantially) faster than in our baseline calibration, the effect of the tariff would be declining in age for the first years, and only increasing after age 20. The reason is that with a fast productivity growth, the optimal capital stock absent adjustment costs is increasing faster than the actual capital stock with adjustment costs.

The V-shaped pattern around age 20 in [Figure 6a](#) is stark and is a direct artifact of how the productivity process is specified. That is, the average productivity continues to grow exponentially up to age 20 and then stops growing all of a sudden. If we would instead assume that productivity growth is a smooth function of age (e.g., such that productivity follows an S-shaped pattern by age), then the effect of a tariff change would also be smoother.

The model is best interpreted as a mechanism exercise rather than a full quantitative accounting of the reduced-form coefficient. It reproduces the sign and age profile of the empirical response, including stronger responses among young firms and a non-

Figure 6: Effect of positive input tariff shock in model.



Notes: The average effects in these graphs are normalized to zero.

monotonicity around middle age. Quantitatively, however, the model generates a smaller differential response than the reduced-form estimates: the age-50 differential is 0.13 in the model compared with 2.08 in [Table 1](#). This partial quantitative fit is consistent with the empirical evidence in [Section 4](#). Including capital intensity as a mediation variable only partially attenuates the age-input tariff interaction, suggesting that adjustment costs in quasi-fixed productive capacity explain part—but not all—of the observed age gradient. This leaves room for other life-cycle margins, such as organizational learning, supplier formation, input-quality upgrading, or demand accumulation, to amplify the quantitative response. We therefore favor a broader interpretation of k as representing slowly adjusting productive capacity, including physical capital as well as other quasi-fixed factors that shape firms' ability to expand over the life cycle.

The capital response provides an untargeted test of the mechanism. The calibration targets life-cycle moments, not the differential investment response to tariff shocks. The model predicts that input liberalization should induce stronger subsequent capital accumulation among young firms because their desired scale rises when variable input costs fall. [Figure 6b](#) shows that this response is indeed strongest for young firms: when input tariffs decline by 100 percentage points, a 5-year-old firm increases its capital stock by 0.09 log points more than a 30-year-old firm. We test this prediction in the data by regressing next year's log capital stock on tariffs interacted with age, controlling for

current capital. Consistent with the model, Table 7 shows that the interaction between age and input tariffs is positive and significant. This pattern supports the interpretation that input liberalization changes firms' dynamic growth paths, rather than only their contemporaneous use of materials.

Table 7: Effect of tariffs on the capital stock next year

	(1)
	$\log k_{t+1}$
τ^{out}	0.0819 (0.0615)
$\tau^{out} \times \text{age}$	-0.00194 (0.00206)
τ^{in}	-0.688*** (0.221)
$\tau^{in} \times \text{age}$	0.0155*** (0.00600)
$\log k_t$	0.634*** (0.0175)
Firm FE	X
Year FE	X
N	12485
R^2	0.739

This table shows the effects of an increase in the tariff by 100 percentage points. Output (resp. input) tariffs at the firm level are weighted averages of the output (resp. input) tariffs of all products the firm produces. Data from PROWESS. This table only includes firms that are between 6 and 50 years old. Robust standard errors are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

The model also clarifies why the age gradient is economically distinct from size or productivity heterogeneity. A young and an old firm with similar current productivity may differ in their dynamic position: the young firm is still moving toward its optimal capital stock, while the old firm is closer to its mature input mix. Input tariff reductions therefore have a larger effect when they arrive early in the life cycle, because they operate on a margin that is still flexible and quantitatively important. This interpretation is consistent with the reduced-form evidence that controls for size and productivity do not eliminate the age interaction.

7 Conclusions

This paper shows that firm age is a distinct determinant of how firms respond to trade liberalization. Using India's trade liberalization during the 1990s, we find that reductions in input tariffs generate substantially larger increases in sales for younger firms than for older firms. These differential responses operate primarily through quantities rather than prices, are not driven by selective survival, and remain after accounting for a broad set of observable firm characteristics, including size, productivity, product scope, trade exposure, financial constraints, and product quality. In contrast, we find no systematic age gradient in firms' responses to reductions in output tariffs. Together, these findings suggest that firm age captures an economically meaningful source of heterogeneity beyond the dimensions typically emphasized in the trade literature.

To interpret these findings, we develop a firm life-cycle model with gradual adjustment of productive capacity, which we capture through physical capital. The model shows that the age gradient arises because firms occupy different positions along the life cycle. Young firms are still building productive capacity and therefore rely more on flexible intermediate inputs, making reductions in input costs especially valuable early in the life cycle. The model reproduces the qualitative age profile observed in the data and generates an additional prediction: input liberalization should induce stronger subsequent capital accumulation among young firms, a prediction that we confirm empirically. More broadly, our results suggest that two firms with similar productivity or size may nevertheless respond differently to trade shocks because they are at different stages of the investment and adjustment process.

Our findings suggest that the effects of trade policy cannot be fully understood without considering where firms are in their life cycle. The absence of a differential age effect from output tariff reductions suggests that young firms are not disproportionately harmed by increased import competition. At the same time, the stronger response of young firms to input tariff reductions implies that input liberalization shifts gains toward the early stages of firm development, potentially raising the value of entry and accelerating the growth of young producers. Whether these dynamic effects translate into larger aggregate welfare

gains depends on equilibrium forces beyond the scope of this paper. More generally, our results highlight firm age as a useful state variable for both empirical analyses and quantitative models of trade.

References

- Akcigit, U., Alp, H., and Peters, M. (2021). Lack of selection and limits to delegation: Firm dynamics in developing countries. *American Economic Review*, 111(1):231–275.
- Bernard, A. B. and Jensen, J. B. (1999). Exceptional exporter performance: Cause, effect, or both? *Journal of International Economics*, 47(1):1–25.
- Bloom, N., Draca, M., and Van Reenen, J. (2016). Trade induced technical change? the impact of chinese imports on innovation, it, and productivity. *The Review of Economic Studies*, 83(1):87–117.
- Caves, R. E. (1998). Industrial organization and new findings on the turnover and mobility of firms. *Journal of Economic Literature*, 36(4):1947–1982.
- Chatterjee, S., Krishna, K., Padmakumar, K., and Zhao, Y. (2025). No country for dying firms: Evidence from india. *NBER Working Paper 33830*.
- Clerides, S. K., Lach, S., and Tybout, J. R. (1998). Is learning by exporting important? micro-dynamic evidence from colombia, mexico, and morocco. *The Quarterly Journal of Economics*, 113(3):903–947.
- Cooper, R. W. and Haltiwanger, J. C. (2006). On the nature of capital adjustment costs. *The Review of Economic Studies*, 73(3):611–633.
- De Loecker, J., Goldberg, P. K., Khandelwal, A. K., and Pavcnik, N. (2016). Prices, markups, and trade reform. *Econometrica*, 84(2):445–510.
- Eaton, J. and Kortum, S. (2002). Technology, geography, and trade. *Econometrica*, 70(5):1741–1779.
- Evans, D. S. (1987). Tests of alternative theories of firm growth. *The Journal of Political Economy*, 95(4):657–674.

- Foster, L., Haltiwanger, J., and Syverson, C. (2008). Reallocation, firm turnover, and efficiency: Selection on productivity or profitability? *American Economic Review*, 98(1):394–425.
- Goldberg, P. K., Khandelwal, A. K., Pavcnik, N., and Topalova, P. (2010). Imported intermediate inputs and domestic product growth: Evidence from india. *The Quarterly Journal of Economics*, 125(4):1727–1767.
- Halpern, L., Koren, M., and Szeidl, A. (2015). Imported inputs and productivity. *American Economic Review*, 105(12):3660–3703.
- Haltiwanger, J., Jarmin, R. S., and Miranda, J. (2013). Who creates jobs? small vs. large vs. young. *The Review of Economics and Statistics*, 95(2):347–361.
- Hasan, R., Mitra, D., and Ramaswamy, K. V. (2007). Trade reforms, labor regulations, and labor-demand elasticities: Empirical evidence from india. *The Review of Economics and Statistics*, 89(3):466–481.
- Hasan, R., Mitra, D., Ranjan, P., and Ahsan, R. N. (2012). Trade liberalization and unemployment: Theory and evidence from india. *Journal of Development Economics*, 97(2):269–280.
- Hopenhayn, H. A. (1992). Entry, exit, and firm dynamics in long run equilibrium. *Econometrica*, 60(5):1127–1150.
- Hsieh, C.-T. and Klenow, P. J. (2014). The life cycle of plants in india and mexico. *The Quarterly Journal of Economics*, 129(3):1035–1084.
- Jovanovic, B. (1982). Selection and the evolution of industry. *Econometrica*, 50(3):649–670.
- Khandelwal, A. K. (2010). The long and short (of) quality ladders. *The Review of Economic Studies*, 77(4):1450–1476.
- Krusell, P., Thürewächter, C., and Weiss, J. (2023). Upwardly mobile: The response of young vs. old firms to monetary policy. *Working Paper*.

- Manova, K. (2013). Credit constraints, heterogeneous firms, and international trade. *The Review of Economic Studies*, 80(2):711–744.
- Melitz, M. J. (2003). The impact of trade on intra-industry reallocations and aggregate industry productivity. *Econometrica*, 71(6):1695–1725.
- Topalova, P. (2010). Factor immobility and regional impacts of trade liberalization: Evidence on poverty from india. *American Economic Journal: Applied Economics*, 2(4):1–41.
- Topalova, P. and Khandelwal, A. (2011). Trade liberalization and firm productivity: The case of india. *Review of Economics and Statistics*, 93(3):995–1009.
- Whited, T. M. and Wu, G. (2006). Financial constraints risk. *Review of Financial Studies*, 19(2):531–559.

Appendix A PROWESS Data

We use data on Indian firms from PROWESS, both at the product level and at the firm level. In addition, we use data on Indian output and input tariffs at the product level in the years 1989–1997 from [De Loecker et al. \(2016\)](#).

Product-level variables we use are sales quantity and its unit of measurement, sales value, and production quantity and its unit of measurement. Sales quantity and sales value are dependent variables in our regressions, while production quantity is used to estimate productivity which is one of our mechanism variables.

We use the year of incorporation to construct the age of the firm. Note that the incorporation year is the year in which the company came into existence as a distinct legal entity, it is not necessarily the year in which the enterprise came into existence. For example, an enterprise could have existed as a partnership or as a proprietorship before it was incorporated as a company. Or the company could be spun-off from an existing firm.

Other firm-level variables we use are total sales, export earnings, labor, materials, capital and the (12-digit 2008) NIC code of the firm’s main economic activity. Our measure of labor is compensation to employees.¹⁴ For capital we use gross fixed assets.¹⁵ And materials are the sum of the expenses incurred on (a) raw materials and on (b) stores, spares and tools consumed. Raw material is the major input for manufacturing companies. Stores and spares aid the production process. These cover sundry supplies, maintenance stores, components, tools, jigs, and other similar equipment.

¹⁴This comprises salaries and wages and social security contributions (e.g., contributions to an insurance company by an employer to pay for medical care of its employees). It includes paid leaves, profit sharing, bonuses and perquisites and non-monetary benefits (such as medical care, housing, cars, and free or subsidised goods or services) for current employees. It also includes post-employment benefits such as gratuity, pension, provident fund and voluntary retirement benefits.

¹⁵This is computed by adding together the historical cost of all fixed assets purchased/constructed, deducting the historical cost of assets sold, and adding or deducting the historical cost of assets coming in or going out at the time of acquisitions, mergers and demergers, etc., as the case may be. This value is thus the sum of gross intangible assets, land & buildings, plant & machinery, computers, electrical installations & fittings, transport infrastructure, transport equipment & vehicles, communication equipment, furniture & fixtures, social amenities and the gross value of any such class of fixed assets.

Cleaning

We first merge all firm-level data. We drop finance firms from our data as we focus on the manufacturing sector for which we have the tariff data. Data are generally reported over a 12-month period. If the data is reported in the first half of the year, we consider the data as pertaining to the previous year. Most of the time, the reporting date is March 31st, as this is the end of the book year in India. If the data is reported over a different length of time than 12 months, we scale all flow variables to a ‘full-year equivalent’.

It sometimes happens that firms provide multiple reports in the same year, although always at different dates. In this case, we select a single observation per firm-year by going through the following steps in this order: (1) keep the observations with the fewest missings on labor, materials and capital; (2) if at least one observation has 12 reporting months while others do not, drop observations with a different number of reporting months; (3) if at least one observation is reported on March 31st, drop observations reported at a different date; and finally, (4) take the first observation of all remaining duplicates.

The product-level data we also convert to full-year equivalents if applicable. Products have a 20-digit CMIE (Centre for Monitoring Indian Economy) code. We drop observations which do not have a product code or whose product code is not in the 20-digit format.¹⁶ We merge with the firm-level data and only keep observations for which the reporting dates of the product-level and firm-level data matches. We then create two datasets for which observations are selected based on similar conditions but for different variables.

The first dataset contains data on sales value, sales quantity, and prices. Prices are not included in the raw data but we construct them from sales values and quantities at the product level. This means that observations for which both sales values and quantities are missing, carry no relevant information, so we drop them. An important thing to note is that not all quantities are reported in the same units of measurement. When

¹⁶The latter consist exclusively of code "10000000000000000000" (1e20) which we suppose means ‘other’ or ‘unknown’.

applicable, we convert units to the same metric unit. This goes for the case of weight, length, area, and volume.

Furthermore, there are two issues we have to deal with: bunching and duplicates. Bunching refers to a situation where one product observation reports the total sales value or quantity for a set of product observations within a firm-year and this data is missing for the other observations in this set. Duplicates refers to a situation in which there are multiple observations of the same product-unit within a firm-year. We deal with these as follows. First, we drop observations which are bunching over different products since there is no way to know the division of the total over the different products. Next, we sum values and quantities over all observations within a firm-year-product-unit set. Finally, we keep only one observation of each duplicate. If, despite the previous steps, an observation does not have data on sales value at this point, we drop it.

The second dataset contains data on production quantities, next to the sales data. We use the production quantities to estimate productivity. In this dataset we use 12-digit product codes by simply removing the last 8 digits of the 20-digit codes. We drop duplicates if none of the observations hold information on one of production quantity, sales quantity, and sales value. Then, we drop duplicate observations for which data on all these variables and the units of measurement are missing. We again convert metric units to a single unit when applicable. There are a number of cases where one duplicate observation holds data on production and another one on sales. We combine these into a single observation if the units are the same and there is no bunching happening.

Lastly, we deal with bunching and duplicates again. We set data on both production and sales to missing when bunching over different product codes occurs. Then, we sum the production and sales data over all observations within a firm-year-product-unit set. Finally, we keep only one observation of each duplicate.

We merge the product-level data from Prowess with the tariff data obtained from [De Loecker et al. \(2016\)](#). The tariff data is reported for an older version of the CMIE product codes, so we map the new codes in our data from PROWESS to these old codes.

Appendix B Results on ASI data

Figure A.1: Histogram of age distribution (1990)-ASI data

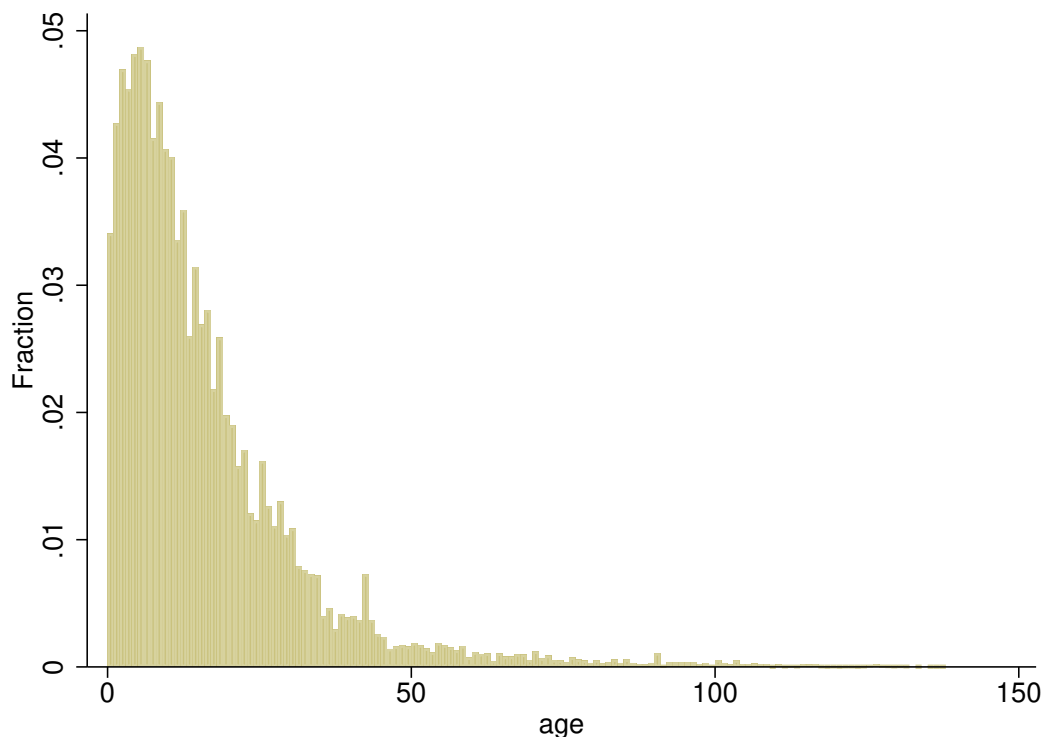


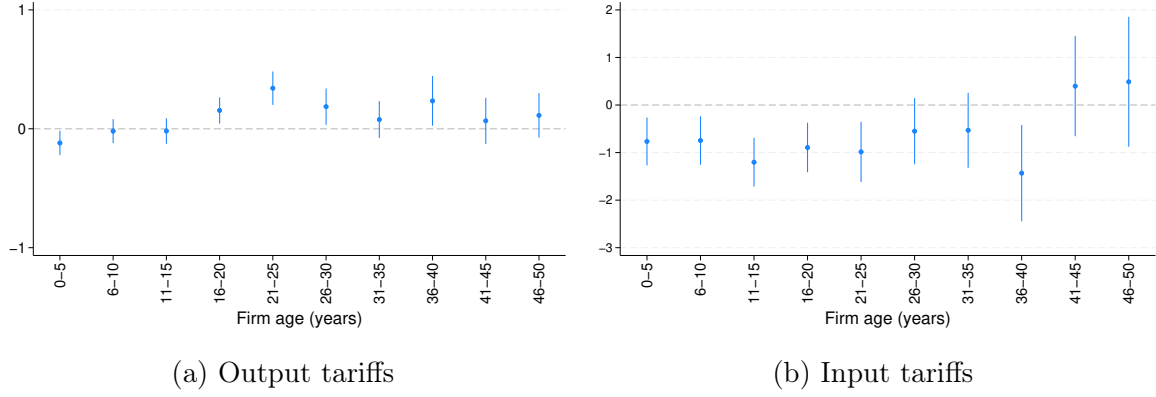
Table A.1: Effect of tariffs on log sales-ASI data

	(1)	(2)	(3)
	s_log	s_log	s_log
τ^{in}	-2.093*** (0.350)	-1.326*** (0.230)	-2.313*** (0.301)
$\tau^{in} \times \text{age}$	0.0209*** (0.00462)	0.0243*** (0.00388)	0.0237*** (0.00387)
Firm sample	5 < age < 50	full	full
Broad Industry-Year FE	X		X
Industry FE	X	X	X
Year FE		X	
N	91996	131725	131725
R^2	0.381	0.387	0.394

Effects of an increase in the tariff by 100 percentage points. Firm and product-level data from ASI. Broad industry and industry are defined at 2 and 4-digit (1987) NIC respectively. Standard errors clustered at the industry level are reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

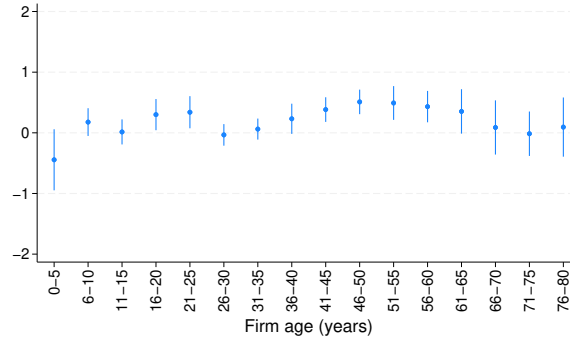
Figure A.2: Differential effect of output tariffs and input tariffs on log sales by firm age-ASI data



Note: Point estimates and 95% confidence intervals for the effect of a change in tariffs on sales for firms of different age bins, as per equation (2). Panel a) shows results for the effect of output tariffs (β_2^b). Panel b) shows results for the effect of input tariffs (β_4^b). Firm-level data from ASI.

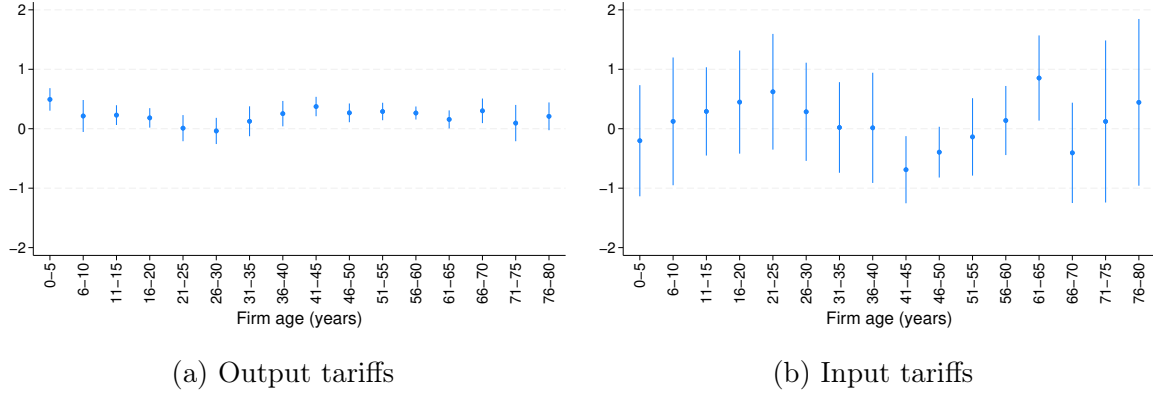
Appendix C Additional Figures

Figure A.3: Differential effect of output tariffs on log sales by firm age



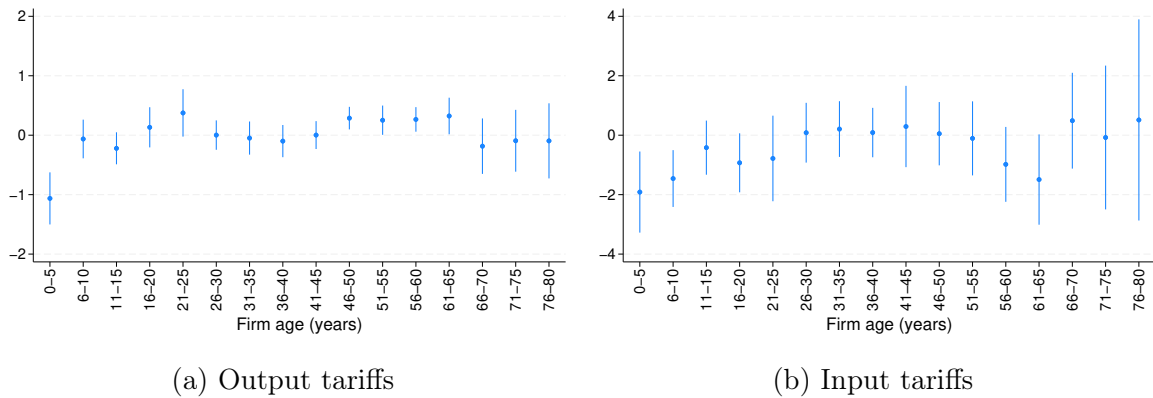
Note: Point estimates and 95% confidence intervals for the effect of a change in tariffs on sales for firms of different age bins, as per equation (2). Firm and product-level data from PROWESS.

Figure A.4: Differential effect of output tariffs and input tariffs on log price by firm age



Note: Point estimates and 95% confidence intervals for the effect of a change in tariffs on sales for firms of different age bins, as per equation (2). Panel a) shows results for the effect of output tariffs (β_2^b). Panel b) shows results for the effect of output tariffs (β_4^b). Firm and product-level data from PROWESS.

Figure A.5: Differential effect of output tariffs and input tariffs on log quantity by firm age



Note: Point estimates and 95% confidence intervals for the effect of a change in tariffs on sales for firms of different age bins, as per equation (2). Panel a) shows results for the effect of output tariffs (β_2^b). Panel b) shows results for the effect of output tariffs (β_4^b). Firm and product-level data from PROWESS.